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GNRO-2017/00016

March 28, 2017

U.S. Nuclear Regulatory Commission
Attn: Document Control Desk
Washington, DC 20555-0001

SUBJECT: Licensee Event Report (LER) 2017-001-00, High Pressure Core Spray Jockey
Pump Trip
Grand Gulf Nuclear Station, Unit 1
Docket No. 50-416
License No. NPF-29

Dear Sir or Madam:

Attached is Licensee Event Report (LER) 2017-001-00, High Pressure Core Spray Jockey Pump Trip. This report is being submitted in accordance with Title 10 Code of Federal Regulations 50.73(a)(2)(v)(D) for any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

This letter contains no new Regulatory Commitments.

If you have any questions or require additional information, please contact James Nadeau at (601)-437-2103.

Sincerely,

JJN/rjf

A handwritten signature in black ink, appearing to read "James Nadeau", written over the typed name "JJN/rjf".

Attachment: Licensee Event Report (LER) 2017-001-00

cc: (See Next Page)

cc: U.S. Nuclear Regulatory Commission
ATTN: Mr. Siva Lingam, NRR/DORL (w/2)
Mail Stop OWFN 8 B1
11555 Rockville Pike
Rockville, MD 20852-2738

U.S. Nuclear Regulatory Commission
ATTN: Mr. Kriss M. Kennedy (w/2)
Regional Administrator, Region IV
1600 East Lamar Boulevard
Arlington, TX 76011-4511

U.S. Nuclear Regulatory Commission
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Regional Administrator, Region IV
1600 East Lamar Boulevard

NRC Senior Resident Inspector
Grand Gulf Nuclear Station
Port Gibson, MS 39150

Attachment to GNRO 2017/00016
Licensee Event Report (LER) 2017-001-00

366

(06-2016)



LICENSEE EVENT REPORT (LER)

(See Page 2 for required number of digits/characters for each block)

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53), U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollections.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOF-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

1. FACILITY NAME Grand Gulf Nuclear Station, Unit 1	2. DOCKET NUMBER 05000416	3. PAGE 1 OF 4
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4. TITLE
High Pressure Core Spray (HPCS) Jockey Pump Trip

5. EVENT DATE			6. LER NUMBER			7. REPORT DATE			8. OTHER FACILITIES INVOLVED	
MONTH	DAY	YEAR	YEAR	SEQUENTIAL NUMBER	REV NO.	MONTH	DAY	YEAR	FACILITY NAME	DOCKET NUMBER
1	27	2017	2017	001	00	3	28	2017	N/A	05000 N/A

9. OPERATING MODE 2	11. THIS REPORT IS SUBMITTED PURSUANT TO THE REQUIREMENTS OF 10 CFR §: (Check all that apply)			
	☐ 20.2201(b)	☐ 20.2203(a)(3)(i)	☐ 50.73(a)(2)(ii)(A)	☐ 50.73(a)(2)(viii)(A)
	☐ 20.2201(d)	☐ 20.2203(a)(3)(ii)	☐ 50.73(a)(2)(ii)(B)	☐ 50.73(a)(2)(viii)(B)
	☐ 20.2203(a)(1)	☐ 20.2203(a)(4)	☐ 50.73(a)(2)(iii)	☐ 50.73(a)(2)(ix)(A)
	☐ 20.2203(a)(2)(i)	☐ 50.36(c)(1)(i)(A)	☐ 50.73(a)(2)(iv)(A)	☐ 50.73(a)(2)(x)
10. POWER LEVEL Raising power in the source range to return to power operations	☐ 20.2203(a)(2)(ii)	☐ 50.36(c)(1)(ii)(A)	☐ 50.73(a)(2)(v)(A)	☐ 73.71(a)(4)
	☐ 20.2203(a)(2)(iii)	☐ 50.36(c)(2)	☐ 50.73(a)(2)(v)(B)	☐ 73.71(a)(5)
	☐ 20.2203(a)(2)(iv)	☐ 50.46(a)(3)(ii)	☐ 50.73(a)(2)(v)(C)	☐ 73.77(a)(1)
	☐ 20.2203(a)(2)(v)	☐ 50.73(a)(2)(i)(A)	☒ 50.73(a)(2)(v)(D)	☐ 73.77(a)(2)(i)
	☐ 20.2203(a)(2)(vi)	☐ 50.73(a)(2)(i)(B)	☐ 50.73(a)(2)(vii)	☐ 73.77(a)(2)(ii)
	☐ 50.73(a)(2)(i)(C)	☐ OTHER	Specify in Abstract below or in NRC Form 366A	

12. LICENSEE CONTACT FOR THIS LER

LICENSEE CONTACT James Nadeau / Manager, Regulatory Assurance	TELEPHONE NUMBER (Include Area Code) (601)-437-2103
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13. COMPLETE ONE LINE FOR EACH COMPONENT FAILURE DESCRIBED IN THIS REPORT

CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX	CAUSE	SYSTEM	COMPONENT	MANU-FACTURER	REPORTABLE TO EPIX
N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

14. SUPPLEMENTAL REPORT EXPECTED ☐ YES (If yes, complete 15. EXPECTED SUBMISSION DATE) ☒ NO	15. EXPECTED SUBMISSION DATE	MONTH	DAY	YEAR
		N/A	N/A	N/A

ABSTRACT (Limit to 1400 spaces, i.e., approximately 15 single-spaced typewritten lines)

At 1808 hours on 1/27/17, Grand Gulf Nuclear Station entered into LCO 3.5.1.B when the High Pressure Core Spray (HPCS) Jockey Pump (Component Function Identifier- P) failed and the HPCS System was declared inoperable. The Reactor Core Isolation Cooling (RCIC) System was verified operable and investigation into the cause was initiated. Under those plant conditions the Plant Technical Specifications action to restore the HPCS System to operable status allows a 14 day completion time.

No other safety systems were inoperable at the time of this event.

The decision was made to disassemble the HPCS Jockey Pump and rebuild the pump using parts from the warehouse which was completed on 1/29/17. The pump was tested to demonstrate functionality of the pump on 1/29/17 and the system was returned to service.

**LICENSEE EVENT REPORT (LER)
CONTINUATION SHEET**

(See NUREG-1022, R.3 for instruction and guidance for completing this form
<http://www.nrc.gov/reading-rm/doc-collections/nuregs/staff/sr1022/r3/>)

Estimated burden per response to comply with this mandatory collection request: 80 hours. Reported lessons learned are incorporated into the licensing process and fed back to industry. Send comments regarding burden estimate to the FOIA, Privacy and Information Collections Branch (T-5 F53); U.S. Nuclear Regulatory Commission, Washington, DC 20555-0001, or by e-mail to Infocollects.Resource@nrc.gov, and to the Desk Officer, Office of Information and Regulatory Affairs, NEOB-10202, (3150-0104), Office of Management and Budget, Washington, DC 20503. If a means used to impose an information collection does not display a currently valid OMB control number, the NRC may not conduct or sponsor, and a person is not required to respond to, the information collection.

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Grand Gulf Nuclear Station, Unit 1	05000-416	YEAR	SEQUENTIAL NUMBER	REV. NO.
		2017	001	00

NARRATIVE**PLANT CONDITIONS PRIOR TO THE EVENT**

At the time of the event, Grand Gulf Nuclear Station (GGNS) Unit 1 was in Mode 2 and raising power in the source range to return to power operations. No other safety systems were inoperable at the time of this event.

DESCRIPTION

At 1808 hours on 1/27/17, the Main Control Room received "HPCS PMP SUCT PRESS ABNORMAL" (1 H13-P601-16A-D4) annunciator due to a High Pressure Core Spray (HPCS) Jockey Pump trip. The Control Room dispatched an operator and electrician to HPCS jockey pump breaker 52-170105 and dispatched an operator locally to the HPCS jockey pump. The operator reported the HPCS breaker to be in the ON position with no abnormalities noted. The local operator reported the HPCS Jockey Pump not running but no other abnormalities were identified. At 1817 hours "HPCS JKY PMP DISCH PRESS LO" annunciator alarmed and "HPCS PMP SUCT PRESS ABNORMAL" annunciator cleared. HPCS pump suction pressure indicated 23.75 psig on Main Control Room Panel 1H13-P625. Average suction pressure over the last five years is 21.3 psig.

The LCO Action Statement entered was LCO 3.5.1.B, ECCS Operating- for HPCS System inoperable.

Initial investigation of the feeder breaker (52-170105) revealed that the overloads at the breaker were tripped. The motor amperage was measured at 53 amps during a 13 second run in which the pump did not rotate. The thermals tripped again. The decision was made to disassemble the HPCS Jockey Pump and rebuild the pump using parts from the warehouse which was performed on 1/29/17. During disassembly of the pump, the following conditions were identified:

- Bad thrust bearing
- A groove in the bearing housing

The pump was tested to demonstrate functionality of the pump on 1/29/17 and the system was returned to service.

REPORTABILITY

Event Notification No. 52516 was made to the U.S. Nuclear Regulatory Commission (NRC) Operations Center. This LER is being submitted pursuant to Title 10 Code of Federal Regulations 10 CFR 50.73(a)(2)(v)(D) for any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident. Telephonic notification was made to the NRC Operations Center via the Emergency Notification System on 1/27/17, at 2305 hours, pursuant to 10 CFR 50.72(b)(3)(v)(D) for any event or condition that could have prevented the fulfillment of the safety function of structures or systems that are needed to mitigate the consequences of an accident.

CAUSE

Direct Cause: The HPCS Jockey Pump failed on 1/27/17 due to the thrust bearing being degraded causing the pump to seize.

Apparent Cause: It was identified that the HPCS Jockey Pump (Gould 3196) does not have a time-based replacement

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CONTINUATION SHEET**

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maintenance task. GGNS implements a predictive monitoring maintenance strategy in lieu of time-based replacement preventive maintenance tasks. The failure of the thrust bearing would have shown up in the axial vibration trends of the pump. Current preventive maintenance was not sufficient to prevent failure of the jockey pump. A conservative time-based replacement PM task to replace the pump will reduce the likelihood of another failure of the pump.

Contributing Cause: The HPCS operating instructions does not contain enough guidance to maintain the HPCS System available when the jockey pump is secured. The inadequate guidance does not cause the jockey pump to fail, but increases the consequences (non-functional time) of the event. The HPCS System can be made operable by keeping the HPCS Pump discharge line full by either the HPCS Jockey Pump or by running the HPCS Pump. During this failure, the procedural guidance did not allow for the HPCS Pump to be run to keep the HPCS System operable.

EXTENT OF CONDITION

The condition may exist in the following equipment, procedures, processes, and/or organizations:

- Residual Heat Removal (RHR) Jockey Pumps A, B, and C
- Low Pressure Core Spray (LPCS) Jockey Pump
- No procedures are affected
- No processes are affected
- No organizations are affected

EXTENT OF CAUSE

The cause may exist in the following equipment, procedures, processes, and/or organizations:

- RHR Jockey Pumps A, B, and C
- LPCS Jockey Pump
- No procedures are affected
- No processes are affected
- No organizations are affected

CORRECTIVE ACTIONS

The HPCS Jockey Pump was rebuilt and a partial HPCS Quarterly Functional Surveillance was performed to demonstrate functionality of the pump. This partial surveillance was completed satisfactorily on 1/29/17 and the pump was returned to service.

To prevent this failure from occurring in the future, the following actions have been taken:

- Initiated preventive maintenance task to perform thermography and oil analysis of the HPCS Jockey Pump bearings.
- Initiated preventive maintenance tasks to perform thermography and oil analysis of all three RHR Jockey Pump bearings.
- Initiated preventive maintenance task to perform thermography and oil analysis of the LPCS Jockey Pump bearings.



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- Initiated preventive maintenance task to replace the HPCS Jockey Pump on a conservative time-based frequency.
- Initiated preventive maintenance task to replace all three RHR Jockey Pumps on a conservative time-based frequency.
- Initiated preventive maintenance task to replace the LPCS Jockey Pump on a conservative time-based frequency.

SAFETY SIGNIFICANCE

The event posed no threat to the health and safety of the general public or to nuclear safety as safety-systems performed as designed. No Technical Specification safety limits were violated. Industrial safety was not challenged, and there was no potential or actual radiological release during the event.

PREVIOUS SIMILAR EVENTS

There are no known previous similar events at Grand Gulf Nuclear Station.